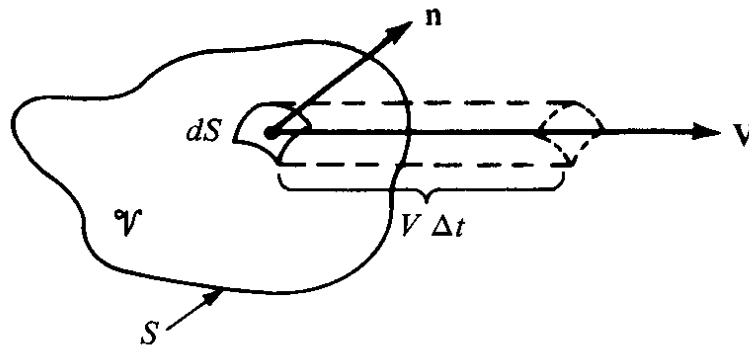


PHYSICAL MEANING OF THE DIVERGENCE OF VELOCITY



the divergence of velocity, $\nabla \cdot \mathbf{V}$

Figure 2.13 Moving control volume used for the physical interpretation of the divergence of velocity.

infinitesimal element of the surface dS moving at the local velocity $\mathbf{V}$
change in the volume of the control volume $\Delta \mathcal{V}$, due to
movement of dS over a time increment Δt ,

$$\Delta \mathcal{V} = [(\mathbf{V} \Delta t) \cdot \mathbf{n}] dS = (\mathbf{V} \Delta t) \cdot \mathbf{dS}$$

PHYSICAL MEANING OF THE DIVERGENCE OF VELOCITY

Over the time increment Δt , the total change in volume of the whole control volume is equal to the summation of Equation (2.28) over the total control surface. In the limit as $dS \rightarrow 0$, the sum becomes the surface integral

$$\oiint_S (\mathbf{V} \Delta t) \cdot d\mathbf{S}$$

If this integral is divided by Δt , the result is physically the time rate of change of the control volume, denoted by DV/Dt ; i.e.,

$$\frac{DV}{Dt} = \frac{1}{\Delta t} \oiint_S (\mathbf{V} \Delta t) \cdot d\mathbf{S} = \oiint_S \mathbf{V} \cdot d\mathbf{S} \quad \mathbf{[2.29]}$$

divergence theorem, Equation (2.26), to the right side of Equation (2.29), we have

$$\frac{DV}{Dt} = \iiint_V (\nabla \cdot \mathbf{V}) dV \quad \mathbf{[2.30]}$$

PHYSICAL MEANING OF THE DIVERGENCE OF VELOCITY

divergence theorem, Equation (2.26), to the right side of Equation (2.29), we have

$$\frac{D\mathcal{V}}{Dt} = \iiint_{\mathcal{V}} (\nabla \cdot \mathbf{V}) d\mathcal{V} \quad \mathbf{[2.30]}$$

Assume that $\delta\mathcal{V}$ is small enough such that $\nabla \cdot \mathbf{V}$ is essentially the same value throughout $\delta\mathcal{V}$. Then the integral in Equation (2.31) can be approximated as $(\nabla \cdot \mathbf{V})\delta\mathcal{V}$. From

$$\frac{D(\delta\mathcal{V})}{Dt} = (\nabla \cdot \mathbf{V})\delta\mathcal{V}$$

$$\boxed{\nabla \cdot \mathbf{V} = \frac{1}{\delta\mathcal{V}} \frac{D(\delta\mathcal{V})}{Dt}}$$

Examine Equation (2.32). It states that $\nabla \cdot \mathbf{V}$ is physically the *time rate of change of the volume of a moving fluid element, per unit volume*. Hence, the interpretation of $\nabla \cdot \mathbf{V}$, first given in Section 2.2.6, Divergence of a Vector Field, is now proved.